

Y13 Half Term 1 – Mechanics Homework (50 marks)

1. Show that $\frac{2x^2-11x-40}{x^2-4x-32} \times \frac{x^2+8x+16}{6x^2-3x-45} \div \frac{8x^2+20x-48}{10x^2-45x+45} = \frac{a}{b}$, and find the value of the constants a and b [2]

2.

The expression $\frac{11+8x}{(2-x)(1+x)^2}$ is denoted by $f(x)$.

- (i) Express $f(x)$ in the form $\frac{A}{2-x} + \frac{B}{1+x} + \frac{C}{(1+x)^2}$, where A , B and C are constants. [5]

3.

Express $\frac{x^2}{(x-1)^2(x-2)}$ in partial fractions. [5]

4.

$$\frac{6-5x-4x^2}{(2-x)(1+2x)} \equiv A + \frac{B}{(2-x)} + \frac{C}{(1+2x)}$$

- (a) Find the values of the constants A , B and C . [4]

5.

Find the quotient and the remainder when $x^4 + 3x^3 + 5x^2 + 4x - 1$ is divided by $x^2 + x + 1$. [4]

6.

Find $\frac{dy}{dx}$ in each of the following cases:

- (i) $y = x^3 e^{2x}$, [2]
(ii) $y = \ln(3 + 2x^2)$, [2]
(iii) $y = \frac{x}{2x+1}$. [2]
(iv) $x^2 \cos 3x$ [2]

(total 8 marks)

7.

Find the equation of the tangent to the curve

$$y = 3x^2(x+2)^6$$

at the point $(-1, 3)$, giving your answer in the form $y = mx + c$.

[5]

8.

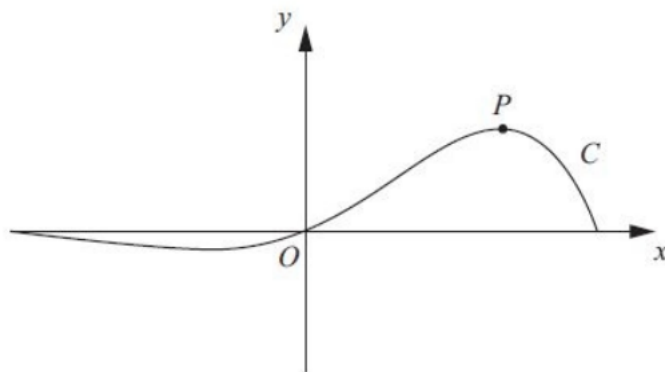


Figure 1

Figure 1 shows a sketch of the curve C which has equation

$$y = e^{x\sqrt{3}} \sin 3x, \quad -\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$$

- (a) Find the x coordinate of the turning point P on C, for which $x > 0$
Give your answer as a multiple of π .
(b) Find an equation of the normal to C at the point where $x = 0$

(6)

(3)

9.

A curve C has equation

$$y = e^{2x} \tan x, \quad x \neq (2n+1)\frac{\pi}{2}.$$

- (a) Show that the turning points on C occur where $\tan x = -1$.
(b) Find an equation of the tangent to C at the point where $x = 0$.

(6)

(2)

Total 50 marks